What Drives the Variation of Developer Communication Characteristics over Time? An Empirical Study Across Multiple Datasets

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Abstract. In large-scale software development ecosystems, the intricate web of developer interactions serves as a framework for cooperative endeavors. This study explores the multifaceted landscape of developer relationships, aiming to uncover the pivotal factors that drive these interactions. We were able to model the dynamic network structures and the evolving patterns of tie formation over time using Temporal Exponential Random Graph Models with Bootstrapped Pseudolikelihood (BTERGMs) in two different software ecosystems. Through meticulous analysis using advanced statistical models, we identified the developer attributes that have the greatest impact on the creation and evolution of these networks. A key novel insight from our analysis is the identification of PageRank as a significant factor shaping developer interactions over time. This discovery has profound implications for individual developers, project managers, and organizational decision-makers, offering valuable insights for navigating the complexities of contemporary software development ecosystems.

Keywords: developer interaction, software development ecosystems, ERGM, BTERGM, PageRank, degree, closeness, temporal dynamics

1 Introduction

In today's software-driven world, the design, development, and maintenance of intricate software systems are crucial. Developers often work in large teams where their interactions facilitate the exchange of information and expertise. As such,

effective engagement among these developers is vital for successful software development. Understanding the factors that drive developer interactions is key to enhancing individual contributions and optimizing team organization and management.

A software development ecosystem can be viewed as a social network, with developers as nodes and connections based on shared interests or collaborations [5]. Models like Erdos Renyi Random Graphs [9], Preferential Attachment Model[3], and Exponential Random Graph Models [14,29] help explain the dynamics of these networks. Applied to large-scale software development, these models highlight how developer interactions drive progress [21].

Traditional analyses of software development networks often offer snapshots of static network structures, leaving nuanced interactions, collaborations, and changes that unfold throughout the software development lifecycle, hidden in obscurity. However, real-life software development networks, comprising developers, interactions, and evolving codebases, are far from static entities; they form dynamic ecosystems that evolve. To gain a deeper understanding of these dynamic networks and the factors that shape their evolution, we analyzed their temporal dynamics.

In this study, we explore the essence of software development networks by employing advanced statistical models tailored to temporal networks. Through this exploration, we zero in on the specific attributes of these nodes, revealing their pivotal roles in shaping collaborative dynamics [32]. In the following two sections, we present a summary of related literature followed by a description of our research. Thereafter we explain our approach and methodologies, discuss our findings, and assess potential risks to their validity. The manuscript concludes with a summary and a list of deductions drawn from the investigation.

2 Literature Review

Network models are frameworks used to represent and analyze the structure and behavior of interconnected systems. Ravasz and Barabasi [24] shed light on the characteristics of real-world networks, including features like heavy-tailed degree distributions, significant clustering tendencies, and the presence of scale-free structures. Dorogovtsev et al. [8] introduced a deterministic network model that maintains the scaling relationship between node degree and clustering coefficient.

Nguyen et al. [22] developed a framework for understanding link formation (LF) behaviors, such as triads and dyads, within dynamic social networks, finding that stability increases as users form more links. Leskovec et al. [17] noticed that real-world networks get denser over time, with links growing faster than the number of nodes, and the average distance between nodes getting shorter. Leung et al. [18] introduced LF-patterns to capture the emergence of new links based on existing connections.

Toral et al. [28] used social network analysis methodologies to conduct an analysis of open-source software development project networks, identifying the

communities responsible for efficient project development, as well as pinpointing the crucial role of brokerage within these communities. Sowe et al. [25] have demonstrated the significance of knowledge brokers within open-source software projects, underscoring their role as expert human resources. Teixeira et al. [27] researched the functions of groups, sub-communities, and business models within the OpenStack ecosystem[5] by employing social network analysis.

Jackson has extensively discussed the concept of Exponential Random Graph Models (ERGMs) in his book titled "Social and Economic Networks" [14]. Additionally, Pol et al. [29] highlighted the growing utilization of ERGMs in the context of social networks due to their capacity to elucidate the global structure of networks while facilitating micro-level tie prediction. Techniques for Maximum Likelihood Estimation (MLE) approximation of ERGMs have been detailed by several researchers [11] [12]. ERGMs have also found utility in exploring longitudinal engagement, performance, and connectivity in social networks [33]. Morris, Handcock, and Hunter [19] outlined methods for controlling the Markov Chain Monte Carlo (MCMC) algorithm that is used for the estimation of ERGM parameters. ERGMs have also been employed to study the structures within large social networks [1]. Wagstrom and Datta [31] applied ERGMs to study developer interactions in globally distributed software development teams and underscored the influences of geographic and temporal separations among the developers on the developer interactions. Bardhan et al. [4] applied ERGMs to Eclipse and OpenStack software development ecosystems and derived closeness centrality to be the driving factor for developer interactions.

Temporal Exponential Random Graph Models (TERGMs) serve as a valuable extension of traditional ERGMs, specifically tailored to capture the dynamic nature of networks over time [15]. TERGMs are notable in dynamic social networks for capturing both static network properties and temporal dependencies, revealing underlying behavioral patterns in network evolution [23]. This modeling approach finds applicability in networks where relationships between nodes are subject to temporal influences, including online social networks, communication networks, and longitudinal data sets capturing human interactions.

BTERGMs [16] are extensions of TERGMs that calculate Maximum Pseudo Likelihood with Bootstrapped Confidence Intervals. This reduces computational demand compared to the modeling algorithm used by traditional TERGMs.

In our research, we utilized the power of BTERGMs to investigate the temporal intricacies of two distinct software development ecosystems. By leveraging BTERGMs, we were equipped to examine and model the shifting network structures and evolving tie formation dynamics within these ecosystems over time.

3 Methodology

In this section, we look into the intricacies of our research methodology outlining the key steps and processes involved. Additionally, a schematic representation

[5] https://www.openstack.org/

of the research methodology has been included in the appendix file linked in the Appendix.

3.1 Overview of the data

In this study, we analyzed data from the OpenStack and Eclipse development ecosystems, shared by Gonzalez-Barahona [10]. OpenStack serves as an open and freely accessible cloud computing platform. Eclipse is an integrated development environment commonly utilized in computer programming which provides a customizable environment. The data provided by Gonzalez-Barahona [10] encompasses details about developer engagements across various interrelated development activities composed of four distinct databases, encompassing source code records from various repositories, problem tickets, various mailing lists, and code reviews.

3.2 Network Construction

Our network construction process centered on identifying developers actively involved in all four core development activities and linking them based on their shared involvement in specific tasks. We constructed co-comment and co-change networks for both OpenStack and Eclipse ecosystems, where the developers are represented as nodes. In co-comment networks, developers were linked if they jointly commented on particular problem tickets, while in co-change networks, links were made between developers who collaborated on specific code modifications. Link weights denoted the frequency of co-commenting or co-changing instances. The statistics of these networks are shown in Table 1. To enhance network clarity, singleton nodes without links were removed.

3.3 Rationale for Metric Selection

We chose six metrics, viz. eigenvector centrality, degree centrality, betweenness centrality, clustering coefficient, closeness centrality, and PageRank. Our choice of network metrics is guided by an approach followed by a previous paper [4], where these metrics were used in an ERGM analysis of fully-formed static networks. In our study, we aim to investigate whether the insights drawn from the ERGM analysis hold true in the context of BTERGMs applied to time-sliced networks. By employing the same set of metrics, we seek to validate and extend the earlier findings, considering the temporal evolution of network dynamics. The metrics, used in our study, are described in the "Overview of the metrics" table, which can be found in the appendix file linked in the Appendix Section.

3.4 Fitting the BTERGM model

ERGM estimates the parameter that best suits the network using MCMC estimation. TERGM is an extension of ERGM modified to work on temporal networks.BTERGM estimates TERGM by maximum pseudo-likelihood estimate

Table 1: Characteristics of the Networks

Network	Nodes	Links	Density	Diameter
Eclipse Changes	420	6467	0.07349699	7
Eclipse Comments	412	4808	0.05678785	7
OpenStack Changes	853	17019	0.04672584	5
OpenStack Comments	826	9390	0.02749223	6

(MPLE) with bootstrapped confidence intervals [16]. The equations of these models are discussed in detail in the appendix file linked in the Appendix section.

Unlike MCMC, MPLE does not require simulation and can be optimized using a simple hill-climbing algorithm. MPLE asymptotically converges to the MLE (Maximum Likelihood Estimation) [26,13] when the number of nodes or sample size increases towards infinity [26,2]. However, MPLE has some drawbacks, such as uncertainty measures based on the MPLE are sometimes severely biased downwards [30]. But this problem can be corrected for TERGMs by bootstrapping the pseudolikelihood [6], which is used in BTERGM.

Thus, BTERGM provides estimates with less computational requirement (when using MPLE instead of MCMC estimate) and can be reliable as it uses bootstrapped pseudolikelihood. This is particularly suitable for large datasets such as ours. For this reason, in this study, we chose BTERGM to estimate the parameters in the co-comment and co-change networks. All networks were fitted using BTERGM for the selected metrics, with 500 bootstrap replications. To speed up the process, the analysis was run in parallel across multiple CPU cores.

4 Results and Discussion

4.1 Model Parameters

The results of the fitting of the BTERGMs to the respective networks are displayed in Tables 2, 3, 4, and 5. These tables provide the parameter estimates along with their corresponding 95% confidence intervals. An estimate is deemed significant if the value 0 does not fall within the calculated confidence interval[16,7]. The parameter estimates provide insights into how likely an edge is to form in the network, based on the presumed predictors. This assists us in understanding if the occurrence of the network is due to the influence of the predictors or purely coincidental. A larger estimate magnitude signifies a more substantial impact of the predictor on network formation. Positive values suggest a favorable influence on link establishment, while negative values indicate an adverse effect [12]. From the tables, PageRank shows the highest positive value, with confidence intervals not including zero, indicating significant and non-random effects.

Metric	Estimate	Boot mean	2.5%	97.5%
Eigen Centrality	-1.0311e+00	-1.0592e+00	-1.2939	-0.9066
Degree	1.9352e-02	1.9374e-02	0.0183	0.0209
Betweenness	-4.3131e-04	-4.3221e-04	-0.0005	-0.0004
Closeness	-2.0653e+01	-2.8968e+01	-85.0433	-6.4973
PageRank	**3.2878e+01**	3.3577e+01	28.9296	40.2620
Cluster Coefficient	-3.2100e+00	-3.1906e+00	-3.2916	-3.0523

Table 2: Eclipse co-change

Metric	Estimate	Boot mean	2.5%	97.5%
Eigen Centrality	-1.2398e+00	-1.2455e+00	-1.8024	-0.7801
Degree	2.1319e-02	2.1434e-02	0.0203	0.0228
Betweenness	-3.7061e-04	-3.7213e-04	-0.0004	-0.0003
Closeness	-1.4051e+02	-1.4929e+02	-292.0031	-45.6336
PageRank	**3.7555e+01**	3.7676e+01	26.5397	51.8384
Cluster Coefficient	-3.3001e+00	-3.2851e+00	-3.4749	-3.0454

Table 3: Eclipse co-comment

Metric	Estimate	Boot mean	2.5%	97.5%
Eigen Centrality	2.0482e-01	2.4022e-01	-0.8580	1.4306
Degree	1.2738e-02	1.2845e-02	0.0116	0.0142
Betweenness	-1.5148e-04	-1.5261e-04	-0.0002	-0.0001
Closeness	-7.1743e+02	-7.3487e+02	-1217.5592	-391.9015
PageRank	**1.9764e+01**	1.9411e+01	14.5856	24.1939
Cluster Coefficient	-3.3903e+00	-3.3678e+00	-3.6058	-3.0717

Table 4: OpenStack co-change

Metric	Estimate	Boot mean	2.5%	97.5%
Eigen Centrality	9.0312e-01	9.0160e-01	0.4058	1.2552
Degree	1.5707e-02	1.5637e-02	0.0135	0.0176
Betweenness	-1.4639e-04	-1.4523e-04	-0.0002	-0.0001
Closeness	-5.7247e+02	-5.9763e+02	-1252.5255	-200.7212
PageRank	**1.7342e+01**	1.7047e+01	4.5911	32.8490
Cluster Coefficient	-4.4054e+00	-4.3682e+00	-4.8629	-3.7390

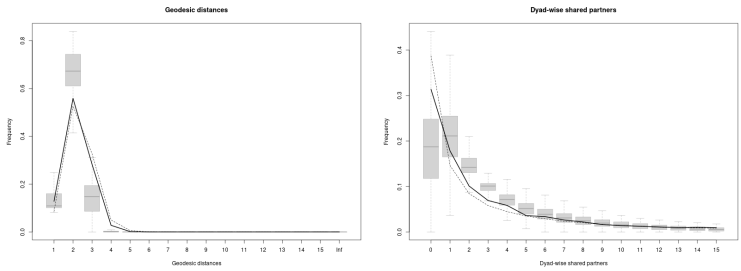
Table 5: OpenStack co-comment

4.2 Evaluating Model Fit

In order to assess how well our statistical model fits the observed temporal network data, we performed a goodness of fit (GOF) analysis of the BTERGM results of the four networks. For this, we can generate multiple networks (in this work, we choose 100 for each network) using the fitted model parameters and then compute various model statistics of the simulated networks to compare to the corresponding statistics of the observed data. For this, we need to choose some network parameters that we did not use to fit the BTERGM models. Here we chose two such properties — geodesic distance and dyad-wise shared partners [20]. The geodesic distance is the smallest number of links that connect two nodes. Dyad-wise shared partners refer to a measure that quantifies the extent to which two nodes (dyad) in a network share connections with other nodes.

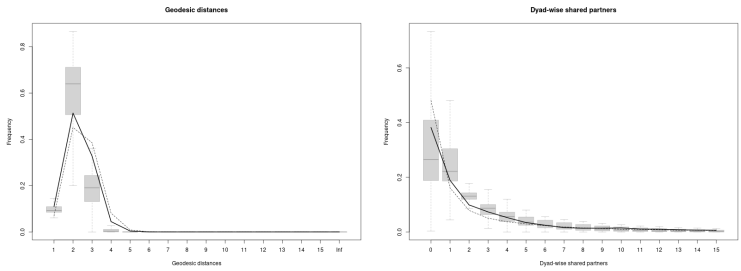
Figures 1, 2, 3, and 4 show the GOF plots of the four networks. In the figures, the solid line represents the statistical values observed in the empirical networks, while the box plots illustrate the averaged statistics derived from the respective models. The higher the proximity of the observed statistics to the averaged statistics from the models, the better the quality of the model's fit to the corresponding data.

In the figures, for all four networks, we observe that the solid lines and the box plots are relatively close to each other, and they become even closer at higher values of the chosen parameters (geodesic distance and dyad-wise shared partners). This indicates a reasonably good fit of the model.



(a) geodesic distances (b) dyad wise shared partners

Fig. 1: GOF of Eclipse co-change

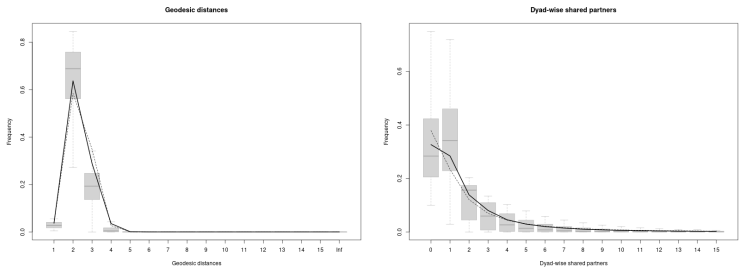


(a) geodesic distances (b) dyad wise shared partners

Fig. 2: GOF of Eclipse co-comment

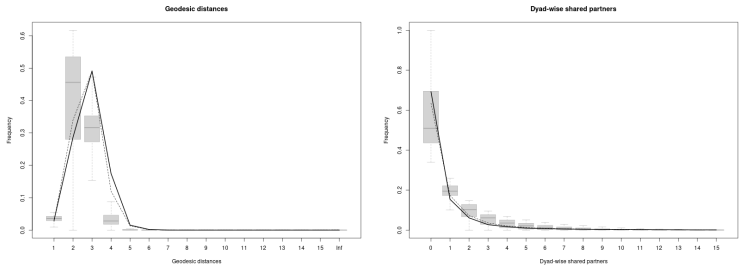
4.3 Observations

It is clear from Tables 2, 3, 4, and 5 that the PageRank metric has emerged as a pivotal factor in the four collaborative development networks that we in-



(a) geodesic distances (b) dyad wise shared partners

Fig. 3: GOF of OpenStack co-change



(a) geodesic distances (b) dyad wise shared partners

Fig. 4: GOF of OpenStack co-comment

vestigated in this study. Previous research [4] emphasized closeness centrality over degree centrality in static networks, with PageRank being the second most influential. However, our time-sliced network analysis reveals that degree centrality has a modest positive influence, while closeness centrality becomes negative. Thus, PageRank emerges as the leading metric in dynamic networks, highlighting a shift from static network views, where closeness centrality was more dominant.

4.4 Implications and Utility

The significance of the network metrics varies with the temporal nature of the network. In static networks, closeness centrality, and PageRank both show high positive values, reflecting strong direct connections and influence. In temporal networks, PageRank remains the most influential metric with a significant positive value, while degree centrality also shows a positive value but with less magnitude. Closeness centrality shifts to a high negative value.

These variations arise from the evolving nature of developer interactions. Pagerank's consistent importance across both static and temporal networks underscores its ability to capture both direct and indirect influence, adapting to changes in developer interactions over time. This persistence highlights its role

in assessing the quality and impact of connections, making it particularly relevant in dynamic contexts. Closeness centrality's shift indicates a move away from direct interactions to more diverse collaborations, while degree centrality's shift reflects increased activity and diversity of connections.

From a utility perspective, PageRank's consistently high values in both static and temporal networks indicate that developers with high PageRank scores exert significant influence. This metric captures the reputation and impact of developers beyond mere connection counts, addressing biases present in degree centrality, which focuses on the number of direct connections. For developers, focusing on relationships with influential peers rather than increasing the number of connections is more effective. Developers with high PageRank value are not only well connected but also have their interactions enhance their influence through implicit endorsements. This insight can guide developers to cultivate strategic connections, improving their collaboration impact and reputation within the network.

5 Threats to validity

While this study has endeavored to shed light on the influential developer attributes within large-scale software development ecosystems, it is important to acknowledge several inherent limitations that may affect the interpretation and generalization of our findings. These limitations are intrinsic to the complexity of the domain and the methodologies employed.

Construct Validity Construct validity concerns the accuracy of our variable measurements. We used standard network metrics, computed consistently across all four networks. While BTERGM was employed for analysis, alternative modeling approaches, such as Stochastic Actor-Oriented Models (SAOMs) or Subgraph Generation Models (SUGMs), could produce different results.

Internal validity Internal validity assesses whether the study accurately identifies cause-and-effect relationships without biases. Our use of rigorously curated historical data from the OpenStack and Eclipse ecosystems enhances the reliability of our measurements over time, minimizing potential biases such as history, maturation, and instrumentation effects.

External validity External validity pertains to the generalizability of our findings beyond the specific context of the study. The results may not apply to all software development ecosystems, as developer interactions can vary widely across different domains and platforms.

Reliability Reliability refers to the degree of consistency, stability, or the capacity to achieve the same results or measurements when the study is replicated under identical conditions. Our findings are reproducible. We have released the code

components produced for this study at https://github.com/SuchandraC/BTERGM to facilitate replication of our results.

6 Future Work

Our future work will concentrate on exploring the broader implications of PageRank's significance, including its relationship with project success, code quality, and community growth. We also want to extend our investigation to diverse software development ecosystems and compare the findings from the OpenStack and Eclipse ecosystems with those from other software development ecosystems to identify common trends or ecosystem-specific factors influencing developer interactions. It will be crucial for assessing the generalizability of our findings. Additionally, we intend to explore the impact of altering the number of time slices in our temporal network analysis. Lastly, we also want to research on finding how other contextual factors may interact with PageRank in shaping developer interactions, providing a more comprehensive view of network dynamics within software development environments.

7 Summary and Conclusions

This study investigates the dynamics of software development collaboration networks by examining co-comment and co-change temporal networks within Eclipse and OpenStack ecosystems. Utilizing BTERGM, we aimed to identify the developer attribute that most significantly influences interactions in these ecosystems. Our findings consistently reveal that PageRank is the most influential metric in both co-comment and co-change networks across both ecosystems. Degree centrality has a minor positive influence, while closeness centrality has a negative impact on network formation. This contrasts with previous research, that highlighted closeness centrality as positively influencing and degree centrality as negatively influencing static network formation [4]. Our results underscore the pivotal role of PageRank in software development ecosystems. Developers with high PageRank scores are central to knowledge flow and collaboration. Unlike other metrics, PageRank focuses on the quality and impact of connections, offering a more accurate measure of influence. These insights are valuable for individual developers and project managers aiming to optimize collaboration. By strategically connecting with influential colleagues, developers can enhance their reputation and elevate the influence of their collaborators within the network.

In summary, our study highlights the centrality of PageRank in understanding software development collaboration networks and provides a strategic approach for thriving in collaborative software development.

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Appendix

The appendix and additional supplementary materials are available at the following GitHub repository for one to download and view:
https://github.com/SuchandraC/BTERGM/blob/main/BTERGM/appendix.pdf.